

NYAH BENTLEY 2026

April 2026 Results

Additional patterns were generated to replicate common spatial light modulator (SLM) outputs, including diffractive structures such as concentric circles and line gratings. These patterns were used to further evaluate the response of the TIE reconstruction under structured illumination.

The resulting reconstructions are consistent with the corresponding in-focus intensity images I_0 , with the imposed patterns clearly reflected in the recovered phase and surface height maps. This agreement indicates that the TIE algorithm effectively captures the spatial distribution of intensity variations introduced by the display.

In the case of the damaged LCD, the defect remains clearly visible in the reconstruction and is further enhanced by the imposed patterns. This reinforces previous observations that TIE reconstruction is highly sensitive to both physical surface features and intensity modulation, with defects appearing more pronounced under structured illumination.

Broken LCD (SLM On)

Stripes and Interference circle waves

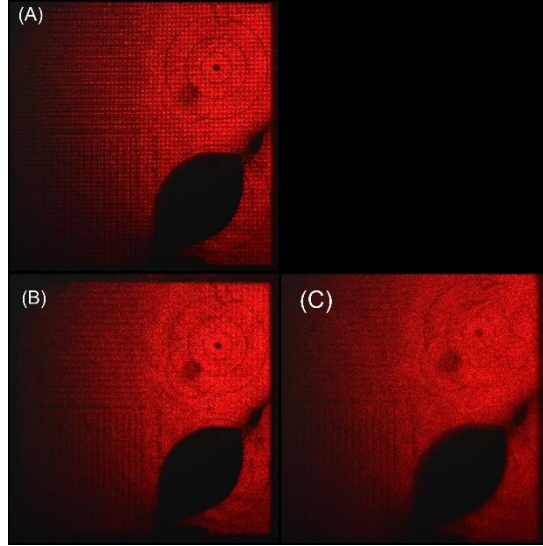


Figure 17 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Stripes and interference circle waves pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction

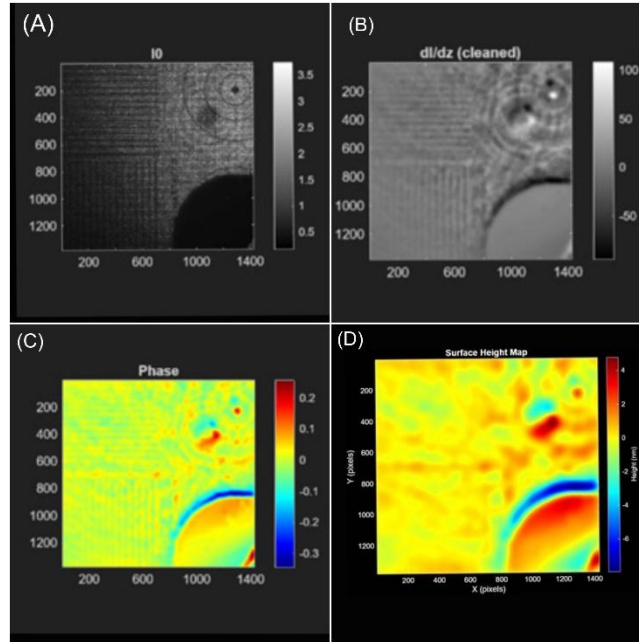


Figure 18 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

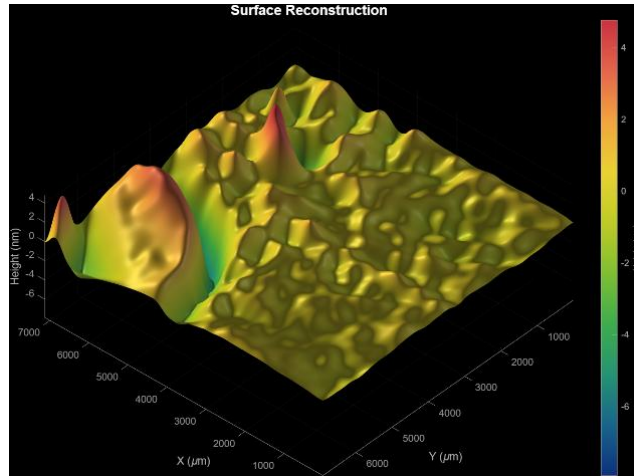


Figure 19 shows the three-dimensional reconstruction of the projected pattern. The damaged region of the LCD appears more pronounced than the imposed checkerboard pattern, indicating that surface defects produce stronger intensity variations and dominate the TIE reconstruction.

Stripes

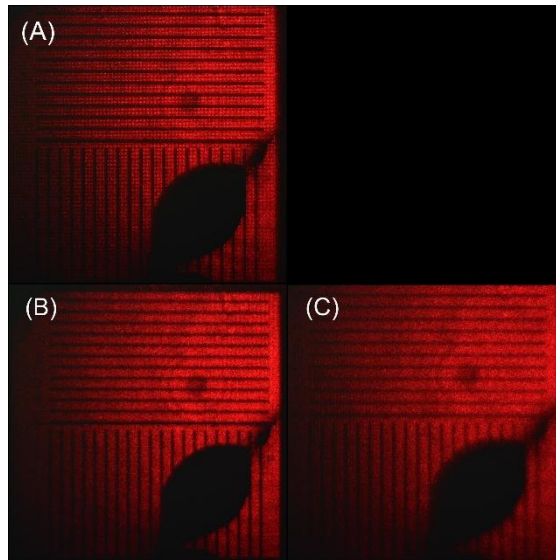


Figure 20 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a Stripes pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction

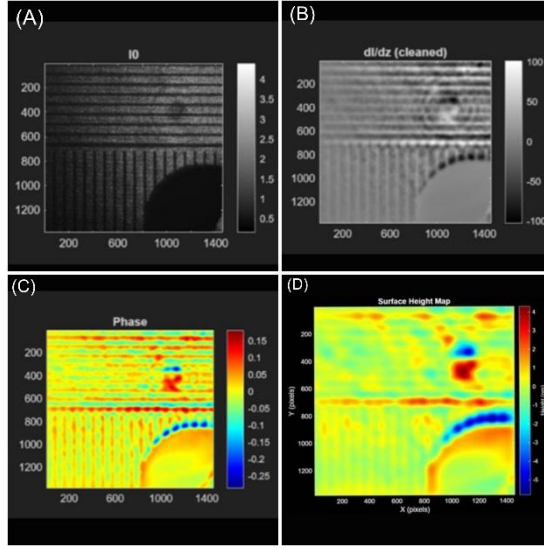


Figure 21 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

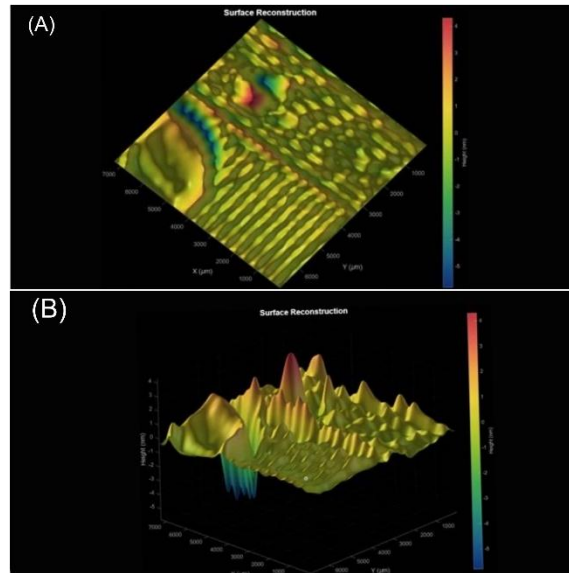


Figure 22 shows the three-dimensional reconstruction of the stripe pattern. The reconstructed features are consistent with the reference image I_0 , with the stripe structure clearly preserved. Additionally, characteristics of the damaged LCD, including surface irregularities, are visible in the reconstruction.

Interference circle waves

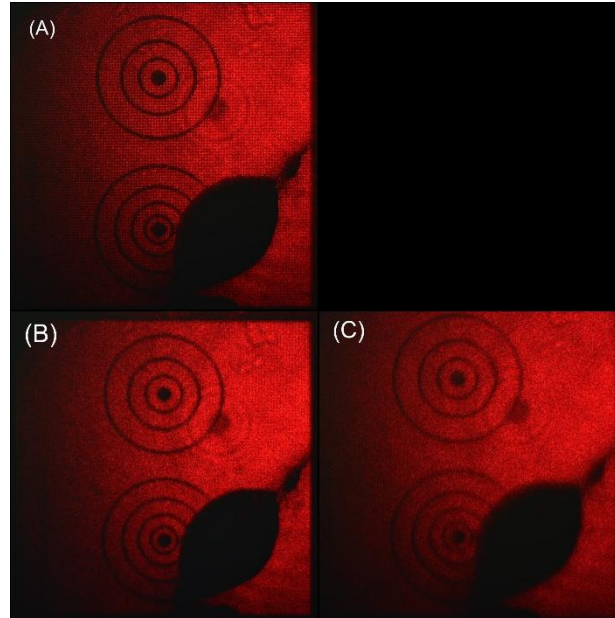


Figure 23 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying an interference circle wave pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction

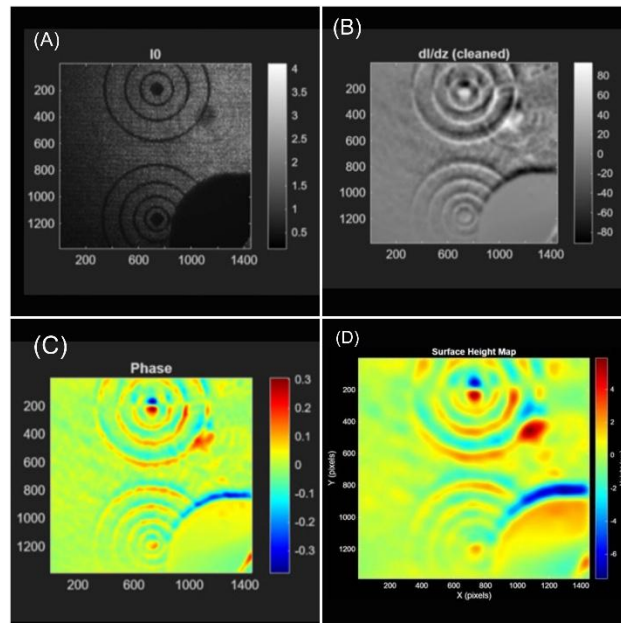


Figure 24 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

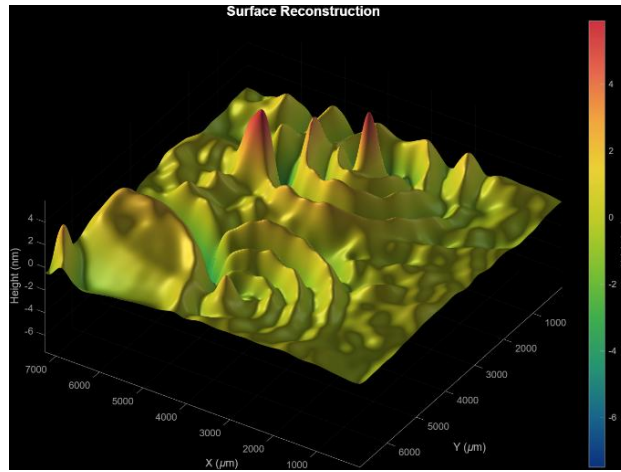


Figure 25 shows the three-dimensional reconstruction of the circular pattern. The reconstructed features closely match the reference image I_0 , with the concentric ring structure clearly preserved. Surface defects in the damaged LCD are also evident, demonstrating that both the imposed pattern and inherent screen irregularities contribute to the reconstructed surface.